Solving the Maximum Diversity Problem Using GRASP and GRASP with Path Relinking

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January 2026

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1 Introduction

The Maximum Diversity Problem (MDP) is a combinatorial optimization problem with numerous practical applications in areas such as facility location, portfolio selection and genetic engineering. Exact solution methods seldom are practical for medium and large instances, making heuristic and metaheuristic approaches more suitable.

In this report, we address the MDP using the Greedy Randomized Adaptive Search Procedure (GRASP) and an enhanced version incorporating Path Relinking. GRASP provides an effective balance between greediness and randomness, while Path Relinking intensifies the search by exploring trajectories that connect high-quality solutions. Both methods are implemented and evaluated on benchmark instances provided in the course and their performance is compared in terms of solution quality and computational time.

2 GRASP

GRASP is a multi-start metaheuristic designed to solve combinatorial optimization problems. Each GRASP iteration consists of two main phases: a construction phase and a local search phase. The best solution found over all iterations is provided as the final result.

2.1 Construction phase

In the construction phase, a feasible solution is built incrementally. At each step, a greedy function is used to evaluate the contribution of adding each candidate element to the current partial solution.

Let $S \subseteq N$ be the current partial solution. For each candidate element $e \in N \setminus S$, its marginal contribution to the objective function is defined as

$$c(e) := \sum_{s \in S} \text{dist}(e, s).$$

Let

$$c^{\max} := \max_{e \in N \setminus S} c(e), \quad c^{\min} := \min_{e \in N \setminus S} c(e).$$

A Restricted Candidate List (RCL) is constructed as

$$\text{RCL} := \left\{ e \in N \setminus S \ : \ c^{\min} \leq c(e) \leq c^{\min} + \alpha(c^{\max} - c^{\min}) \right\}$$

where $\alpha \in [0, 1]$ is a parameter that controls the level of randomness in the construction phase.

An element is randomly selected from the RCL and added to S. This process is repeated until S is completed.

2.2 Local Search phase

Once a complete solution is constructed, a local search procedure is applied to improve it. The neighborhood is defined by swap moves: replacing a selected element with a non-selected one.

Given a solution S, a neighboring solution S' is obtained as

$$S' := (S \setminus \{s\}) \cup \{e\}, \quad s \in S,\ e \in N \setminus S.$$

The change in the objective function produced by this move is

$$\sum_{k \in S \setminus \{s\}} (\text{dist}(e, k) - \text{dist}(s, k)).$$

The local search iteratively applies improving moves until no further improvement is possible, reaching a local optimum.

3 GRASP with Path Relinking

GRASP with Path Relinking (GRASP+PR) is an extension of the standard GRASP algorithm that incorporates an intensification strategy. The main idea is to explore trajectories that connect high-quality solutions in order to discover better solutions.

3.1 Elite Set management

During the execution of GRASP, a set of elite solutions E is maintained. This elite set contains high-quality and sufficiently diverse solutions obtained from different GRASP iterations.

A new generated solution S is added to E if its objective value is better than that at least one solution in E and it is sufficiently different from the existing elite solutions.

The size of the elite set is fixed and determined through parameter calibration.

3.2 Path Relinking procedure

Given two elite solutions S_{I} (initiating solution) and S_{G} (guiding solution), Path Relinking generates a sequence of intermediate solutions that gradually transform S_{I} into S_{G}.

Let

$$A := S_{\mathrm{G}} \setminus S_{\mathrm{I}}, \quad B := S_{\mathrm{I}} \setminus S_{\mathrm{G}}.$$

At each step, an element $b \in B$ is replaced by an element $a \in A$, generating a new solution

$$S' := (S_{\mathrm{I}} \setminus \{b\}) \cup \{a\}.$$

In Greedy Path Relinking, all possible such exchanges are evaluated and the move producing the best objective value is selected.

The selected move is applied and the process continues until $S_{\mathrm{I}} = S_{\mathrm{G}}$. The best solution found along the path is retained.

Overall, the static version of GRASP with PR consists of these steps:

1. Generate multiple solutions using GRASP.
2. Select elite solutions based on quality and diversity.
3. Apply Path Relinking between each pair of elite solutions.
4. Keep the best solution found.

3.3 Code structure of GRASP with PR

The project is structured around the following modules (some modules have been previously presented in the course and they have not been modified):

- `instance.py`.
- `solution.py`.
- `cgrasp.py`.
- `lsbestimp.py`.
- `grasp.py`.

- `elite.py`: Manages the elite set used in the Path Relinking. It maintains a fixed-size list of high-quality and diverse solutions obtained from the GRASP phase, ensuring no duplicates and keeping the best solutions sorted by objective value.
- `path_relinking.py`: Implements the Path Relinking procedure. Given an initiating and a guiding solution from the elite set, it generates a path by performing the best possible swap moves that transform the initiating solution into the guiding one. The best solution found along this trajectory is returned.
- `grasp_pr.py`: Implements GRASP with Path Relinking. First, runs a GRASP phase to construct an elite set. Next, it applies Path Relinking between pairs of elite solutions, followed by local search as an improvement method.

3.4 Parameters calibration

Two parameters are required to be calibered: the scalar α of GRASP and the number of elements on the elite group.

After some systematic experimentation fixing the random seed and solving the 15 given instances for each $\alpha \in \{0.1, 0.2, \ldots, 0.8, 0.9\}$ and an elite size in $\{5, 10, 15\}$, we have not appreciated a significant change on the results of our implementation.

For that reason, we have decided to consider an α and an elite size reasonable in theory. From now on, we will set $\alpha = 0.4$ and an elite size of 10. In that way, we will balance greediness ($\alpha \approx 0$) and randomness ($\alpha \approx 1$). Besides, an elite size of 10 may permit some diversity but will not require too much computational effort.

4 GRASP and GRASP+PR comparison

For each of the 15 instances, both algorithms were executed with a time limit of 30 seconds and a fixed seed 1. The results obtained were the following (we show the value of the objective function of all instances and the solutions to only the small instances for convenience):

Instance	**GRASP[sol]**	**GRASP+PR[sol]**
MDG-a_1_100_m10	1 37 74 43 13 80 86 57 26 31	1 37 74 43 13 80 86 57 26 31
MDG-a_2_500_m50	-	-
MDG-a_4_100_m10	1 34 2 38 71 44 45 47 90 29	1 2 34 38 71 44 45 47 90 29
MDG-a_5_500_m50	-	-
MDG-a_6_500_m50	-	-
MDG-a_9_500_m50	-	-
MDG-a_10_100_m10	34 68 69 73 74 12 14 51 52 90	34 68 69 73 74 12 14 51 52 90
MDG-a_12_100_m100	64 13 15 20 86 93 96 51 55 59	64 96 13 15 51 20 86 55 59 93
MDG-a_13_500_m50	-	-
MDG-a_14_100_m10	65 67 72 51 53 30 22 25 94 95	65 67 72 51 53 22 30 25 94 95
MDG-a_16_500_m50	-	-
MDG-a_19_500_m50	-	-
MDG-a_20_100_m10	67 77 82 85 25 92 93 40 52 53	67 40 77 82 52 53 85 25 92 93
MDG-a_20_500_m50	-	-

Instance	**GRASP[of]**	**GRASP+PR[of]**
MDG-a_1_100_m10	360.15	360.15
MDG-a_2_500_m50	**7718.82**	7691.04
MDG-a_4_100_m10	355.72	355.72
MDG-a_5_500_m50	7683.38	**7691.83**
MDG-a_6_500_m50	7702.68	**7723.35**
MDG-a_9_500_m50	**7738.91**	7726.96
MDG-a_10_100_m10	355.5	355.5
MDG-a_12_100_m100	354.25	354.25
MDG-a_13_500_m50	**7724.38**	7701.99
MDG-a_14_100_m10	356.06	356.06
MDG-a_16_500_m50	7726.03	**7760.28**
MDG-a_17_500_m50	**7710.36**	7666.74
MDG-a_19_500_m50	**7710.1**	7686.56
MDG-a_20_100_m10	349.31	349.31
MDG-a_20_500_m50	**7683.38**	7623.52

As the results show, none method is consistently better than the other. On one hand, small instances produce approximately the same objective

value. On the other hand, there are some discrepancies for larger instances. The main reason we can identify looking at the code trace (see python console when running main.py) is that, for larger instances, GRASP with PR is incapable of either filling the elite size or completing the PR in 30 seconds. Normally, it is limited to a elite group of five solutions and two iterations of PR.

After some experimentation, we have determined that duplicating the time limit (60 seconds) does not affect GRASP objective values and only improves the GRASP with PR ones. Nonetheless, the solution it produces is seldom significantly better.

5 Summary and observations

In this work, the Maximum Diversity Problem has been addressed using two metaheuristic approaches: the standard GRASP algorithm and an enhanced version incorporating Path Relinking. Both methods were implemented and tested on the benchmark instances provided in the course, under identical computational conditions and a fixed time limit.

The computational results show that, for small instances, both GRASP and GRASP with Path Relinking obtain very similar objective values. In many of these cases, both methods converge to the same solution, indicating that the basic GRASP procedure is already capable of finding high-quality solutions within a short time.

For larger instances, some discrepancies between the two methods can be observed. While GRASP with Path Relinking occasionally produces better solutions, it does not consistently outperform standard GRASP. A further analysis of the execution reveals that the additional computational effort required by Path Relinking limits its effectiveness under a strict time constraint. In particular, for large instances, the algorithm is often unable to fully populate the elite set or to perform an extensive Path Relinking phase within the thirty second limit.

Overall, these observations suggest that Path Relinking is a powerful intensification strategy, but its benefits strongly depend on the available computational time and the size of the instances. When time is limited, standard GRASP offers a very competitive performance with a lower computational cost. On the other hand, when more time is available, GRASP with Path Relinking can further refine solutions and occasionally achieve better objective

values.

It is noted that our implementation could be improved by assuring more diversity on the elite set and by improving its efficiency.